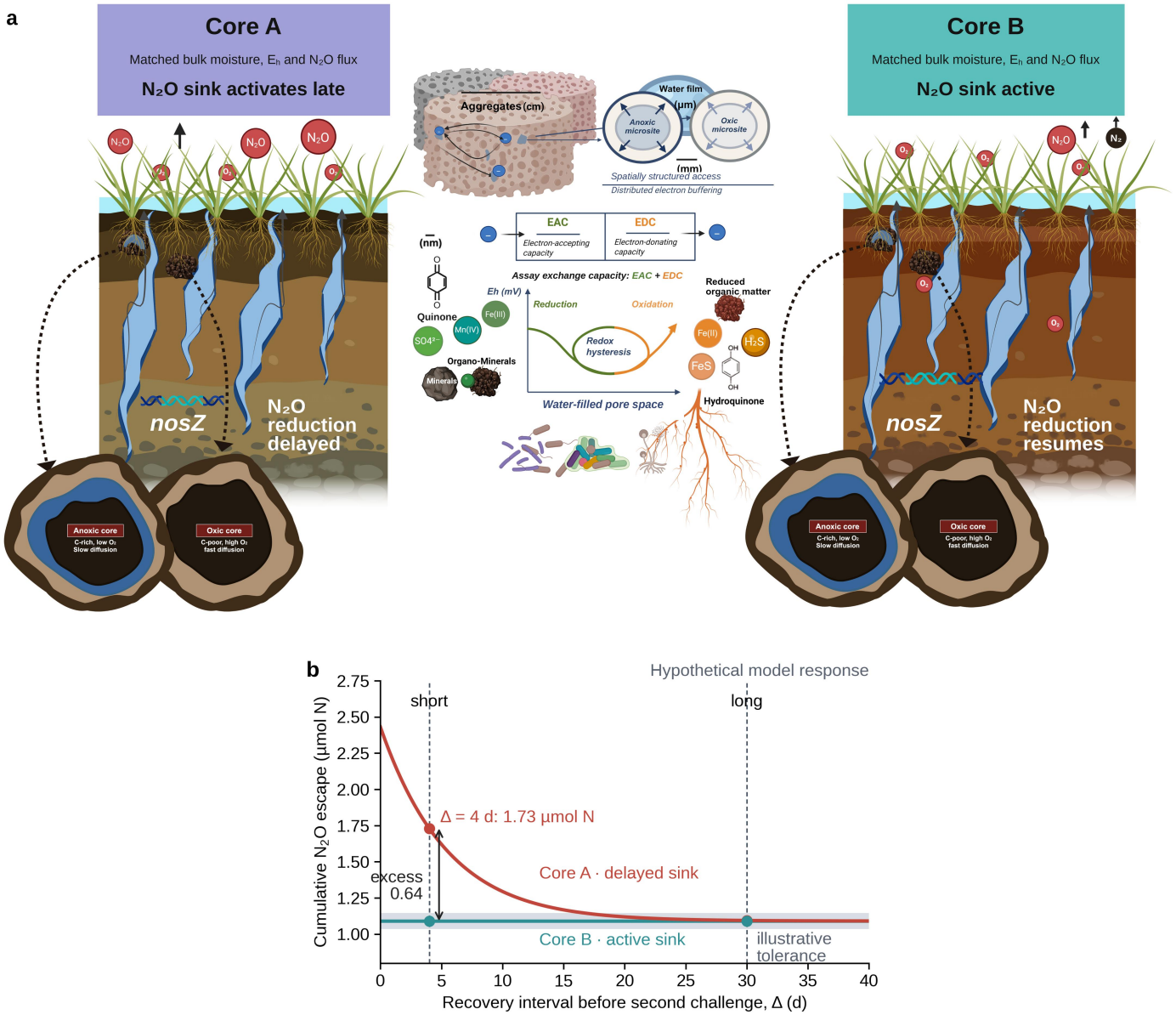

a
Core A
Matched bulk moisture, Eh and N2O flux
N2O sink activates late
Core B
Matched bulk moisture, Eh and N2O flux
N2O sink active
Aggregates (cm)
Water film (µm)
Anoxic microsite
Oxic microsite
(mm)
Spatially structured access
Distributed electron buffering
(nm)
EAC
Electron-accepting capacity
EDC
Electron-donating capacity
Assay exchange capacity: EAC + EDC
Eh (mV)
Reduction
Oxidation
Redox hysteresis
Quinone
Fe(III)
Mn(IV)
SO4 2-
Organo-Minerals
Minerals
Reduced organic matter
Fe(II)
H2S
FeS
Hydroquinone
Water-filled pore space
nosZ
N2O reduction delayed
nosZ
N2O reduction resumes
Anoxic core
C-rich, low O2 Slow diffusion
Oxic core
C-poor, high O2 fast diffusion
Anoxic core
C-rich, low O2 Slow diffusion
Oxic core
C-poor, high O2 fast diffusion
b
Hypothetical model response
short
long
Δ = 4 d: 1.73 µmol N
excess 0.64
Core A · delayed sink
Core B · active sink
illustrative tolerance
Cumulative N2O escape (µmol N)
Recovery interval before second challenge, Δ (d)
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0
5
10
15
20
25
30
35
40